

Grouped Summary Function Demo

Julia Gómez-Coronado

Purpose of the Function

This project demonstrates a custom R function called `grouped_summary()`.

The function summarizes a numeric variable across groups within a dataset. It returns the variable name, number of observations, number and percentage of missing values, and descriptive statistics including the mean, median, standard deviation, minimum, and maximum.

This function is useful for exploratory data analysis because it allows the user to compare data quality and variable patterns across categories or repeated time periods in one clear table.

Setup

```
library(tidyverse)
library(knitr)
```

Load Data

This demo uses R's built-in `airquality` dataset, which contains daily air quality measurements in New York from May to September 1973.

```
data("airquality")
names(airquality)
```

```
[1] "Ozone"    "Solar.R" "Wind"     "Temp"     "Month"    "Day"
```

```
str(airquality)
```

```
'data.frame': 153 obs. of 6 variables:
 $ Ozone : int 41 36 12 18 NA 28 23 19 8 NA ...
 $ Solar.R: int 190 118 149 313 NA NA 299 99 19 194 ...
 $ Wind : num 7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
 $ Temp : int 67 72 74 62 56 66 65 59 61 69 ...
 $ Month : int 5 5 5 5 5 5 5 5 5 5 ...
 $ Day : int 1 2 3 4 5 6 7 8 9 10 ...
```

Custom Function

```
grouped_summary <- function(x, group, var_name) {
  tibble(
    Month = group,
    Value = x
  ) |>
  group_by(Month) |>
  summarise(
    Variable = var_name,
    Observations = n(),
    Missing = sum(is.na(Value)),
    Missing_Pct = round(sum(is.na(Value)) / n() * 100, 1),
    Mean = round(mean(Value, na.rm = TRUE), 1),
    Median = round(median(Value, na.rm = TRUE), 1),
    SD = round(sd(Value, na.rm = TRUE), 1),
    Min = min(Value, na.rm = TRUE),
    Max = max(Value, na.rm = TRUE),
    Check = ifelse(sum(is.na(Value)) > 0, "Missing values", "Complete"),
    .groups = "drop"
  )
}
```

Function Output

I applied `grouped_summary()` to three numeric variables in the `airquality` dataset: `Ozone`, `Solar.R`, and `Temp`. Each variable is summarized by month.

```
air_quality <- bind_rows(
  grouped_summary(airquality$Ozone, airquality$Month, "Ozone"),
  grouped_summary(airquality$Solar.R, airquality$Month, "Solar.R"),
  grouped_summary(airquality$Temp, airquality$Month, "Temp")
)
```

)

```
kable(air_quality)
```

Month	Vari- able	Observa- tions	Miss- ing	Miss- ing_Pct	Mean	Me- dian	SD	Min	Max	Check
5	Ozone	31	5	16.1	23.6	18.0	22.2	1	115	Missing values
6	Ozone	30	21	70.0	29.4	23.0	18.2	12	71	Missing values
7	Ozone	31	5	16.1	59.1	60.0	31.6	7	135	Missing values
8	Ozone	31	5	16.1	60.0	52.0	39.7	9	168	Missing values
9	Ozone	30	1	3.3	31.4	23.0	24.1	7	96	Missing values
5	So- lar.R	31	4	12.9	181.3	194.0	115.1	8	334	Missing values
6	So- lar.R	30	0	0.0	190.2	188.5	92.9	31	332	Complete
7	So- lar.R	31	0	0.0	216.5	253.0	80.6	7	314	Complete
8	So- lar.R	31	3	9.7	171.9	197.5	76.8	24	273	Missing values
9	So- lar.R	30	0	0.0	167.4	192.0	79.1	14	259	Complete
5	Temp	31	0	0.0	65.5	66.0	6.9	56	81	Complete
6	Temp	30	0	0.0	79.1	78.0	6.6	65	93	Complete
7	Temp	31	0	0.0	83.9	84.0	4.3	73	92	Complete
8	Temp	31	0	0.0	84.0	82.0	6.6	72	97	Complete
9	Temp	30	0	0.0	76.9	76.0	8.4	63	93	Complete

Interpretation

The output shows monthly observations, missing values, percentage of missing values, and descriptive statistics for each variable. For example, **Ozone** has missing values in several months, while **Temp** is complete across all months.

The function makes it easier to compare both data quality and variable patterns across repeated time periods. This is useful before deeper analysis because missingness and distributional differences can affect interpretation, model quality, and visualization choices.